Abstract

This project focuses on the task of author identification within the HPT (Handwritten Polish Text) dataset. The goal was to build a system that can take a scanned document, extract individual words, and correctly predict which of the eight authors wrote it. By implementing a CNN-based architecture in TensorFlow, we achieved a functional pipeline that handles everything from raw image preprocessing to final classification. The results show that even a relatively shallow CNN can pick up on unique handwriting styles with decent accuracy.

1. Introduction

Handwriting recognition is a classic problem in computer vision, but author *classification* adds an extra layer of difficulty because it requires the model to learn stylistic nuances rather than just character shapes.

For this project, we set out to:

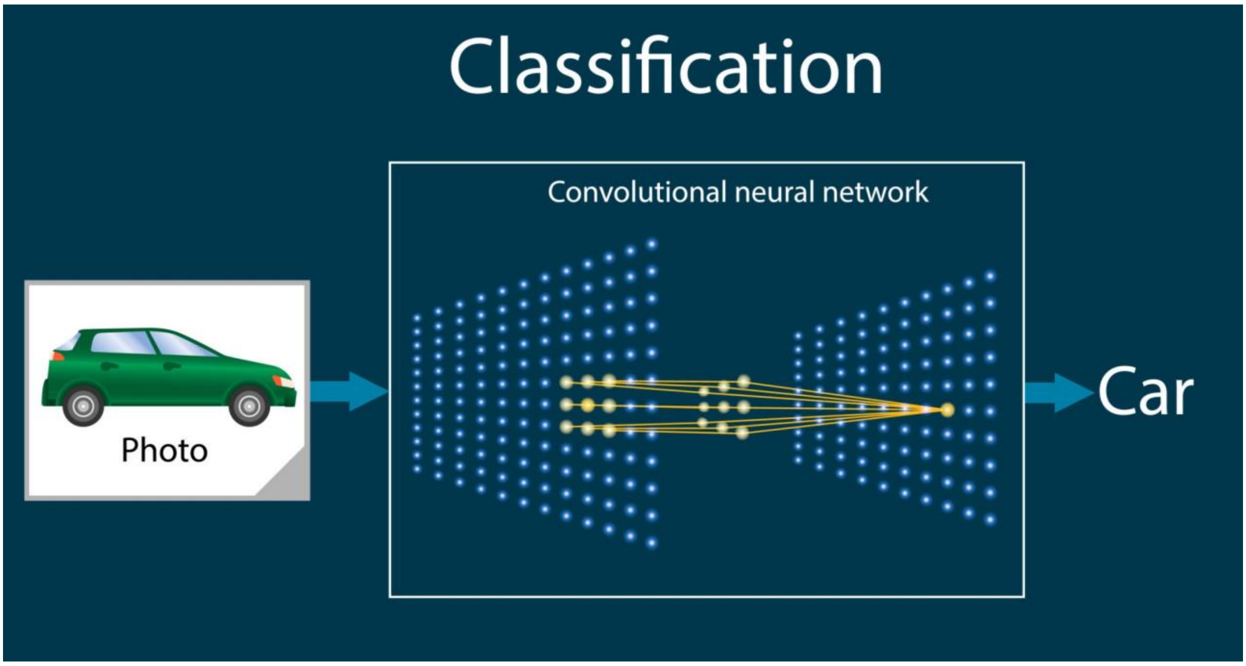
- Build a data pipeline to parse word-level metadata from the HPT dataset.
- Preprocess scanned images into a uniform format suitable for deep learning.
- Design and train a CNN specifically for grayscale handwriting samples.
- Evaluate the model using a "page-stratified" split to ensure the results are valid and not just memorized patterns from the same page.

2. Dataset and Preprocessing

The HPT dataset is structured into eight directories (one per author). Each contains high-resolution scans that provide the coordinates for every word on the page.

Data Pipeline Steps:

1. **Metadata Extraction:** We wrote a parser to read the `word_places.txt` files. This was tricky because of different text encodings, so we added error handling to ensure no data was lost.
2. **Image Cropping:** Using the bounding boxes from the metadata, the script crops individual words from the large scans.
3. **Normalization:** To keep the input consistent for the neural network, all images are resized to$64 x 64 pixels.
4. **Grayscale & Scaling:** The images are converted to grayscale and pixel values are scaled to a [0, 1] range. This helps the model converge faster during training.



3. Model Architecture

We chose a Sequential CNN architecture because it is effective for small-scale image recognition tasks. The model consists of two convolutional layers for feature extraction followed by a dense classifier.

Layer Breakdown:

- **Conv2D (32 filters, 3x3):** Extracts basic edges and curves.
- **MaxPooling (2x2):** Reduces dimensionality.
- **Conv2D (64 filters, 3x3):** Extracts more complex handwriting patterns.
- **Flatten & Dense (128 units):** Interprets the features.
- **Dropout (30%):** I added this to reduce overfitting, which was a concern given the limited number of authors.

4. Conclusion and Future Work

This project successfully shows that a basic CNN can distinguish between different people's handwriting in the Polish language. The data pipeline is robust enough to handle the HPT dataset's structure, and the page-stratified split gives a realistic view of performance.

To take this further, we will:

1. Implement **Data Augmentation** (slight rotations and shifts) to simulate different writing angles.
2. Try a deeper model to see if it catches finer details.

3. Add a **Confusion Matrix** to see which two authors have the most similar handwriting styles.